

Bit Manipulation.

- (i) Check if i th bit set
($a \& (1 \ll i)$)
- (ii) Set i th bit
($a | (1 \ll i)$)
- (iii) Toggle i th bit
($a \wedge (1 \ll i)$)
- (iv) ($a \& \sim(1 \ll i)$) $\rightarrow$ Unset i th bit
- (v) $n \& (\sim(n-1)) \rightarrow$ ~~LSB~~ get LSB

* Bitset

- 1.) if k times others, 1 element 1 time
find that element.
($\% k$)

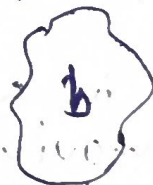
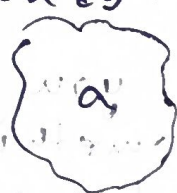
- 2.) if others appear twice, 2 elements appear once, find those 2 elements.
($a \oplus b$)

idea $\rightarrow$ separate them.

s.t. $a \& b$ are in diff. cat.

now apply logic to

bits to them.



Say $n \neq a, b$

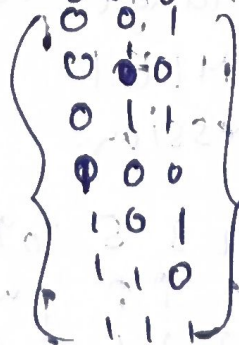
if $n \in \text{grp}(a)$

then all ~~the~~ duplicates of n must also belong to $\text{grp}(a)$.

3.) Techniques / Blueprints

- (i) Cyclic property of bits

eg. to calc. no. of 1's from 1 to n



We can observe

that i th bit is 2^{i-1} periodic.

$\therefore$ whenever series like $n=1, 2, 3, \dots$ given $\rightarrow$ think of this

* (ii) Contribution Techniques:-

Whenever bit expressions (eg. $a \& b + c \& d$...)

given, think of which terms in the expression

will have their i th bit (2^i) set, & then calculate

(Bit expressions are sum independent on each bit)

(ii) Conservation of bits.

$a+b = a \& b + a | b$ (think of them as applying gravity)

suppose $\sum a_i = \text{const.}$

$\therefore \sum q_i^2$ is maximized if applied gravity,

greedy idea.

eg $a+b=5$

$b=5 \quad a=0$

$b=3 \quad a=2$

$b=4 \quad a=1$

$\rightarrow (a+b)^2 =$

$a^2 + b^2 = 25 (\text{max})$

$a^2 + b^2 = 13 (\text{min})$

$\rightarrow 17.$

(iv) Highest to lowest

in max/min type of q's

we this $\underline{1000} > \underline{0111}$.

(MSB matters the most.)

(v) to get range queries, we might use prefix sums & then apply above ideas.